

The Definition of the Derivative

ANSWER KEY



Using the limit definition

- 1 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 3x^2 - x$.

$$\frac{3(x+h)^2 - (x+h) - (3x^2 - x)}{h} = \frac{6xh + 3h^2 - h}{h} = 6x + 3h - 1 \rightarrow 6x - 1.$$

- 2 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+2}$.

$$\frac{\frac{1}{x+h+2} - \frac{1}{x+2}}{h} = \frac{(x+2) - (x+h+2)}{h(x+h+2)(x+2)} = \frac{-h}{h(x+h+2)(x+2)} = \frac{-1}{(x+h+2)(x+2)} \rightarrow \frac{-1}{(x+2)^2}.$$

The derivative as a limit at a point

- 3 Use $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ to find $f'(3)$ for $f(x) = x^2 - 2x$.

$$f(3+h) = (3+h)^2 - 2(3+h) = 9 + 6h + h^2 - 6 - 2h = 3 + 4h + h^2. f(3) = 3. \frac{4h+h^2}{h} = 4 + h \rightarrow 4: f'(3) = 4.$$

- 4 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3$.

$$\frac{x^3 - 8}{x - 2} = \frac{(x-2)(x^2 + 2x + 4)}{x - 2} = x^2 + 2x + 4 \rightarrow 4 + 4 + 4 = 12: f'(2) = 12.$$

Differentiability and graphical interpretation

- 5 Explain why $f(x) = |x - 2|$ is not differentiable at $x = 2$, computing the one-sided derivatives.

Left-hand: $\lim_{h \rightarrow 0^-} \frac{|h|}{h} = -1$. Right-hand: $\lim_{h \rightarrow 0^+} \frac{|h|}{h} = 1$. Since these disagree, $f'(2)$ does not exist (the graph has a corner).

- 6 State three conditions under which a function fails to be differentiable at a point, with a brief description of each (e.g. corner, cusp, vertical tangent, discontinuity).

(1) A corner — one-sided derivatives disagree, as with $|x|$. (2) A vertical tangent line — the slope becomes infinite, as with $\sqrt[3]{x}$ at $x = 0$. (3) A discontinuity — if f isn't continuous at a point, it can't be differentiable there, since differentiability implies continuity.

Power, sum, and constant rules

- 7 Differentiate each function using the power rule:

$$\text{a } f(x) = 7x^5 = 35x^4 \quad \text{b } f(x) = \frac{4}{x^2} = -8x^{-3} \quad \text{c } f(x) = 6\sqrt[3]{x} = 2x^{-2/3} \quad \text{d } f(x) = 9 = 0$$

- 8 Differentiate $f(x) = 5x^4 - 3x^3 + 2x - 7$.

$$f'(x) = 20x^3 - 9x^2 + 2.$$

Tangent lines from the derivative

- 9 Using the limit definition result $f'(x) = 6x - 1$ found earlier for $f(x) = 3x^2 - x$, find the equation of the tangent line to f at $x = 1$.

$$f(1) = 3 - 1 = 2; f'(1) = 6 - 1 = 5. \text{ Tangent line: } y - 2 = 5(x - 1), \text{ i.e. } y = 5x - 3.$$

- 10 Find the equation of the tangent line to $f(x) = x^3$ at $x = 2$, using $f'(2) = 12$ (found earlier).

$$f(2) = 8. \text{ Tangent line: } y - 8 = 12(x - 2), \text{ i.e. } y = 12x - 16.$$

Estimating derivatives numerically

- 11 Given $f(1.9) = 3.42$ and $f(2.1) = 3.98$, use a symmetric difference quotient to estimate $f'(2)$.

$$f'(2) \approx \frac{f(2.1) - f(1.9)}{2.1 - 1.9} = \frac{3.98 - 3.42}{0.2} = \frac{0.56}{0.2} = 2.8.$$

Visualizing the derivative as a tangent slope

- 12 The graph shows $f(x) = 3x^2 - x$ (from an earlier problem) with its tangent line at $x = 1$, where $f'(x) = 6x - 1$.

a Confirm the tangent line's slope matches $f'(1)$. $f'(1) = 6(1) - 1 = 5$, matching the tangent line $y = 5x - 3$, which has slope 5.

b Explain what it means graphically for the line to be "tangent" at this point. Being tangent means the line touches the curve at $x = 1$ and has the same instantaneous rate of change (slope) as the curve at that exact point, so locally the line is the best straight-line approximation to the curve there.

More practice with the limit definition

- 13 Use the limit definition to find $f'(x)$ for $f(x) = \sqrt{x+1}$.

$$\frac{\sqrt{x+h+1} - \sqrt{x+1}}{h} = \frac{(x+h+1) - (x+1)}{h(\sqrt{x+h+1} + \sqrt{x+1})} = \frac{h}{h(\sqrt{x+h+1} + \sqrt{x+1})} = \frac{1}{\sqrt{x+h+1} + \sqrt{x+1}} \rightarrow \frac{1}{2\sqrt{x+1}}.$$

The derivative in applied motion contexts

- 14 A ball's height is $h(t) = -5t^2 + 30t + 1$ (meters, seconds). Use the limit definition of the derivative (or the power rule, since it agrees) to find $h'(t)$, then find the ball's velocity at $t = 2$.

$h'(t) = -10t + 30$. At $t = 2$: $h'(2) = -20 + 30 = 10$ m/s (the ball is still rising).

- 15 For the same height function, find the time at which the ball's velocity is 0 (its highest point).

$h'(t) = -10t + 30 = 0$ gives $t = 3$ s.

Comparing average and instantaneous rate of change

- 16 For $f(x) = x^2$, find the average rate of change on $[1, 3]$, then compare it to the instantaneous rate of change $f'(2)$ at the midpoint $x = 2$.

Average rate: $\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4$. Instantaneous rate: $f'(x) = 2x$, so $f'(2) = 4$. They match exactly here, because for a quadratic, the average rate over an interval always equals the instantaneous rate at the interval's midpoint.

Multiple choice — easy 15

- 17 For $f(x) = 3x^2 - 4x - 1$, use the power rule to find $f'(2)$.

- a. 2
- b. 6
- c. 8 ✓
- d. 10

$f'(x) = 6x - 4$, so $f'(2) = 6(2) - 4 = 8$.

- 18 For $f(x) = 5x^2 - 5x + 4$, use the power rule to find $f'(3)$.

- a. 10
- b. 25 ✓
- c. 27
- d. 23

$f'(x) = 10x - 5$, so $f'(3) = 10(3) - 5 = 25$.

19 For $f(x) = 3x^2 + 4x + 6$, use the power rule to find $f'(-1)$.

- a. -4
- b. 0
- c. 1
- d. -2 ✓

$$f'(x) = 6x + 4, \text{ so } f'(-1) = 6(-1) + 4 = -2.$$

20 For $f(x) = 4x^2 - 8x - 4$, use the power rule to find $f'(-2)$.

- a. -22
- b. -16
- c. -24 ✓
- d. -26

$$f'(x) = 8x - 8, \text{ so } f'(-2) = 8(-2) - 8 = -24.$$

21 For $f(x) = 5x^2 - 4x - 2$, use the power rule to find $f'(2)$.

- a. 6
- b. 16 ✓
- c. 18
- d. 14

$$f'(x) = 10x - 4, \text{ so } f'(2) = 10(2) - 4 = 16.$$

22 For $f(x) = x^2 - 9x + 3$, use the power rule to find $f'(-3)$.

- a. -12
- b. -13
- c. -17
- d. -15 ✓

$$f'(x) = 2x - 9, \text{ so } f'(-3) = 2(-3) - 9 = -15.$$

23 For $f(x) = 2x^2 - 6x - 9$, use the power rule to find $f'(3)$.

a. 6 ✓

b. 8

c. 0

d. 4

$$f'(x) = 4x - 6, \text{ so } f'(3) = 4(3) - 6 = 6.$$

24 For $f(x) = 6x^2 + 9x - 5$, use the power rule to find $f'(3)$.

a. 43

b. 27

c. 45 ✓

d. 47

$$f'(x) = 12x + 9, \text{ so } f'(3) = 12(3) + 9 = 45.$$

25 For $f(x) = 6x^2 + 4x + 4$, use the power rule to find $f'(0)$.

a. 4 ✓

b. 6

c. 2

d. 5

$$f'(x) = 12x + 4, \text{ so } f'(0) = 12(0) + 4 = 4.$$

26 For $f(x) = 3x^2 + 3x + 4$, use the power rule to find $f'(-4)$.

a. -21 ✓

b. -19

c. -23

d. -9

$$f'(x) = 6x + 3, \text{ so } f'(-4) = 6(-4) + 3 = -21.$$

27 For $f(x) = 5x^2 + 4x + 4$, use the power rule to find $f'(0)$.

a. 6

b. 5

c. 2

d. 4 ✓

$$f'(x) = 10x + 4, \text{ so } f'(0) = 10(0) + 4 = 4.$$

28 For $f(x) = 3x^2 - 9x + 3$, use the power rule to find $f'(-4)$.

a. -31

b. -35

c. -33 ✓

d. -21

$$f'(x) = 6x - 9, \text{ so } f'(-4) = 6(-4) - 9 = -33.$$

29 For $f(x) = 5x^2 + 3x - 6$, use the power rule to find $f'(0)$.

a. 5

b. 1

c. 3 ✓

d. 4

$$f'(x) = 10x + 3, \text{ so } f'(0) = 10(0) + 3 = 3.$$

30 For $f(x) = x^2$, use the power rule to find $f'(3)$.

a. 4

b. 3

c. 6 ✓

d. 8

$$f'(x) = 2x, \text{ so } f'(3) = 2(3) = 6.$$

31 For $f(x) = 2x^2 - 6x - 2$, use the power rule to find $f'(-2)$.

a. -14 ✓

b. -12

c. -16

d. -10

$$f'(x) = 4x - 6, \text{ so } f'(-2) = 4(-2) - 6 = -14.$$

Multiple choice — medium 15

32 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 - 2x + 5$.

a. $2x - 3$

b. $x - 2$

c. $2x - 2$ ✓

d. $2x + 2$

$$\frac{1(x+h)^2 - 2(x+h) - (x^2 - 2x + 5)}{h} = \frac{2xh + 1h^2 - 2h}{h} = 2x + 1h - 2 \rightarrow 2x - 2.$$

33 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 - 6x + 8$.

a. $2x - 7$

b. $2x - 6$ ✓

c. $x - 6$

d. $2x + 6$

$$\frac{1(x+h)^2 - 6(x+h) - (x^2 - 6x + 8)}{h} = \frac{2xh + 1h^2 - 6h}{h} = 2x + 1h - 6 \rightarrow 2x - 6.$$

34 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 - 6x$.

a. $x - 6$

b. $2x - 7$

c. $2x - 6$ ✓

d. $2x + 6$

$$\frac{1(x+h)^2 - 6(x+h) - (x^2 - 6x)}{h} = \frac{2xh + 1h^2 - 6h}{h} = 2x + 1h - 6 \rightarrow 2x - 6.$$

35 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 2x^2 + x + 4$.

a. $2x + 1$

b. $4x - 1$

c. $4x + 1$ ✓

d. $4x + 2$

$$\frac{2(x+h)^2 + (x+h) - (2x^2 + x + 4)}{h} = \frac{4xh + 2h^2 + h}{h} = 4x + 2h + 1 \rightarrow 4x + 1.$$

36 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 2x^2 - x - 4$.

a. $2x - 1$

b. $4x - 1$ ✓

c. $4x + 1$

d. $4x - 2$

$$\frac{2(x+h)^2 - (x+h) - (2x^2 - x - 4)}{h} = \frac{4xh + 2h^2 - h}{h} = 4x + 2h - 1 \rightarrow 4x - 1.$$

37 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 6x^2 + 3x + 2$.

a. $6x + 3$

b. $12x - 3$

c. $12x + 3$ ✓

d. $12x + 4$

$$\frac{6(x+h)^2 + 3(x+h) - (6x^2 + 3x + 2)}{h} = \frac{12xh + 6h^2 + 3h}{h} = 12x + 6h + 3 \rightarrow 12x + 3.$$

38 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 4x^2 - 3x + 2$.

a. $8x + 3$

b. $4x - 3$

c. $8x - 4$

d. $8x - 3$ ✓

$$\frac{4(x+h)^2 - 3(x+h) - (4x^2 - 3x + 2)}{h} = \frac{8xh + 4h^2 - 3h}{h} = 8x + 4h - 3 \rightarrow 8x - 3.$$

39 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 4x^2 + 9x - 7$.

a. $4x + 9$

b. $8x + 9$ ✓

c. $8x - 9$

d. $8x + 10$

$$\frac{4(x+h)^2 + 9(x+h) - (4x^2 + 9x - 7)}{h} = \frac{8xh + 4h^2 + 9h}{h} = 8x + 4h + 9 \rightarrow 8x + 9.$$

40 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 - 5x$.

a. $2x - 6$

b. $2x - 5$ ✓

c. $2x + 5$

d. $x - 5$

$$\frac{1(x+h)^2 - 5(x+h) - (x^2 - 5x)}{h} = \frac{2xh + 1h^2 - 5h}{h} = 2x + 1h - 5 \rightarrow 2x - 5.$$

41 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 6x^2 + x + 9$.

a. $12x + 2$

b. $12x + 1$ ✓

c. $6x + 1$

d. $12x - 1$

$$\frac{6(x+h)^2 + (x+h) - (6x^2 + x + 9)}{h} = \frac{12xh + 6h^2 + h}{h} = 12x + 6h + 1 \rightarrow 12x + 1.$$

42 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 5x^2 - 4x - 9$.

a. $10x + 4$

b. $10x - 4$ ✓

c. $5x - 4$

d. $10x - 5$

$$\frac{5(x+h)^2 - 4(x+h) - (5x^2 - 4x - 9)}{h} = \frac{10xh + 5h^2 - 4h}{h} = 10x + 5h - 4 \rightarrow 10x - 4.$$

43 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 5x^2 - 9$.

a. $15x$

b. $10x + 1$

c. $10x$ ✓

d. $5x$

$$\frac{5(x+h)^2 - (5x^2 - 9)}{h} = \frac{10xh + 5h^2}{h} = 10x + 5h \rightarrow 10x.$$

44 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 - 5x - 6$.

- a. $x - 5$
- b. $2x - 6$
- c. $2x + 5$
- d. $2x - 5$ ✓

$$\frac{1(x+h)^2 - 5(x+h) - (x^2 - 5x - 6)}{h} = \frac{2xh + 1h^2 - 5h}{h} = 2x + 1h - 5 \rightarrow 2x - 5.$$

45 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 3x^2 - 7x + 2$.

- a. $6x + 7$
- b. $6x - 7$ ✓
- c. $6x - 8$
- d. $3x - 7$

$$\frac{3(x+h)^2 - 7(x+h) - (3x^2 - 7x + 2)}{h} = \frac{6xh + 3h^2 - 7h}{h} = 6x + 3h - 7 \rightarrow 6x - 7.$$

46 Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = 6x^2 - 2x + 4$.

- a. $12x - 2$ ✓
- b. $6x - 2$
- c. $12x + 2$
- d. $12x - 3$

$$\frac{6(x+h)^2 - 2(x+h) - (6x^2 - 2x + 4)}{h} = \frac{12xh + 6h^2 - 2h}{h} = 12x + 6h - 2 \rightarrow 12x - 2.$$

Multiple choice — hard 15

47 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+6}$, then evaluate $f'(0)$.

a. $-\frac{1}{6}$

b. $-\frac{1}{36}$ ✓

c. $-\frac{1}{37}$

d. $\frac{1}{36}$

$$f'(x) = \frac{-1}{(x+6)^2}, \text{ so } f'(0) = \frac{-1}{36} = -\frac{1}{36}.$$

48 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+6}$, then evaluate $f'(4)$.

a. $-\frac{1}{100}$ ✓

b. $-\frac{1}{10}$

c. $\frac{1}{100}$

d. $-\frac{1}{101}$

$$f'(x) = \frac{-1}{(x+6)^2}, \text{ so } f'(4) = \frac{-1}{100} = -\frac{1}{100}.$$

49 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+1}$, then evaluate $f'(-3)$.

a. $\frac{1}{2}$

b. $-\frac{1}{5}$

c. $-\frac{1}{4}$ ✓

d. $\frac{1}{4}$

$$f'(x) = \frac{-1}{(x+1)^2}, \text{ so } f'(-3) = \frac{-1}{4} = -\frac{1}{4}.$$

50 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+5}$, then evaluate $f'(-2)$.

a. $-\frac{1}{10}$

b. $-\frac{1}{9}$ ✓

c. $\frac{1}{9}$

d. $-\frac{1}{3}$

$$f'(x) = \frac{-1}{(x+5)^2}, \text{ so } f'(-2) = \frac{-1}{9} = -\frac{1}{9}.$$

51 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x+2}$, then evaluate $f'(2)$.

- a. $-\frac{1}{4}$
- b. $\frac{1}{16}$
- c. $-\frac{1}{17}$
- d. $-\frac{1}{16}$ ✓

$$f'(x) = \frac{-1}{(x+2)^2}, \text{ so } f'(2) = \frac{-1}{16} = -\frac{1}{16}.$$

52 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(-2)$ for $f(x) = x^3 + 6x$.

- a. 6
- b. 0
- c. 20
- d. 18 ✓

$$\frac{f(x) - f(-2)}{x - (-2)} = \frac{x^3 - (-2)^3 + 6(x - (-2))}{x - (-2)} = x^2 + (-2)x + (-2)^2 + 6 \rightarrow 3(-2)^2 + 6 = 18.$$

53 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(4)$ for $f(x) = x^3 - 3x$.

- a. 51
- b. 45 ✓
- c. 47
- d. 9

$$\frac{f(x) - f(4)}{x - (4)} = \frac{x^3 - (4)^3 - 3(x - (4))}{x - (4)} = x^2 + (4)x + (4)^2 - 3 \rightarrow 3(4)^2 - 3 = 45.$$

54 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(3)$ for $f(x) = x^3 - 2x$.

a. 29

b. 25 ✓

c. 27

d. 7

$$\frac{f(x) - f(3)}{x - (3)} = \frac{x^3 - (3)^3 - 2(x - (3))}{x - (3)} = x^2 + (3)x + (3)^2 - 2 \rightarrow 3(3)^2 - 2 = 25.$$

55 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(-2)$ for $f(x) = x^3 - 4x$.

a. 10

b. 8 ✓

c. -10

d. 16

$$\frac{f(x) - f(-2)}{x - (-2)} = \frac{x^3 - (-2)^3 - 4(x - (-2))}{x - (-2)} = x^2 + (-2)x + (-2)^2 - 4 \rightarrow 3(-2)^2 - 4 = 8.$$

56 Use the alternate form $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find $f'(-3)$ for $f(x) = x^3 + 3x$.

a. 32

b. 24

c. 30 ✓

d. -6

$$\frac{f(x) - f(-3)}{x - (-3)} = \frac{x^3 - (-3)^3 + 3(x - (-3))}{x - (-3)} = x^2 + (-3)x + (-3)^2 + 3 \rightarrow 3(-3)^2 + 3 = 30.$$

57 For $f(x) = x^2$ when $x \leq 0$, and $f(x) = 5x$ when $x > 0$ (chosen so f is continuous at $x = 0$), determine whether f is differentiable at $x = 0$.

a. the left-hand derivative is $2(0) = 0$ and the right-hand derivative is 5 — they agree, so f is differentiable at $x = 0$.

b. f is differentiable at $x = 0$ automatically, since it is continuous there.

c. the left-hand derivative is $2(0) = 0$ and the right-hand derivative is 5 — they disagree, but differentiability doesn't require one-sided derivatives to match.

d. the left-hand derivative is $2(0) = 0$ and the right-hand derivative is 5 — they disagree, so f is not differentiable at $x = 0$ (the graph has a corner). ✓

Continuity is given. For differentiability, compare one-sided derivatives: the left-hand derivative is $2(0) = 0$ and the right-hand derivative is 5 — they disagree, so f is not differentiable at $x = 0$ (the graph has a corner).

58 For $f(x) = x^2$ when $x \leq 3$, and $f(x) = 5x - 6$ when $x > 3$ (chosen so f is continuous at $x = 3$), determine whether f is differentiable at $x = 3$.

a. f is differentiable at $x = 3$ automatically, since it is continuous there.

b. the left-hand derivative is $2(3) = 6$ and the right-hand derivative is 5 — they disagree, so f is not differentiable at $x = 3$ (the graph has a corner). ✓

c. the left-hand derivative is $2(3) = 6$ and the right-hand derivative is 5 — they agree, so f is differentiable at $x = 3$.

d. the left-hand derivative is $2(3) = 6$ and the right-hand derivative is 5 — they disagree, but differentiability doesn't require one-sided derivatives to match.

Continuity is given. For differentiability, compare one-sided derivatives: the left-hand derivative is $2(3) = 6$ and the right-hand derivative is 5 — they disagree, so f is not differentiable at $x = 3$ (the graph has a corner).

59 For $f(x) = x^2$ when $x \leq 2$, and $f(x) = x + 2$ when $x > 2$ (chosen so f is continuous at $x = 2$), determine whether f is differentiable at $x = 2$.

a. f is differentiable at $x = 2$ automatically, since it is continuous there.

b. the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 1 — they agree, so f is differentiable at $x = 2$.

c. the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 1 — they disagree, so f is not differentiable at $x = 2$ (the graph has a corner). ✓

d. the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 1 — they disagree, but differentiability doesn't require one-sided derivatives to match.

Continuity is given. For differentiability, compare one-sided derivatives: the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 1 — they disagree, so f is not differentiable at $x = 2$ (the graph has a corner).

60 For $f(x) = x^2$ when $x \leq -1$, and $f(x) = -2x - 1$ when $x > -1$ (chosen so f is continuous at $x = -1$), determine whether f is differentiable at $x = -1$.

a. the left-hand derivative is $2(-1) = -2$ and the right-hand derivative is -2 — they disagree, so f is not differentiable at $x = -1$.

b. the left-hand derivative is $2(-1) = -2$ and the right-hand derivative is -2 — they agree, so f is differentiable at $x = -1$. ✓

c. the left-hand derivative is $2(-1) = -2$ and the right-hand derivative is -2 — they agree, but f is not differentiable since the pieces use different formulas.

d. f is differentiable at $x = -1$ automatically, since it is continuous there.

Continuity is given. For differentiability, compare one-sided derivatives: the left-hand derivative is $2(-1) = -2$ and the right-hand derivative is -2 — they agree, so f is differentiable at $x = -1$.

- 61 For $f(x) = x^2$ when $x \leq 2$, and $f(x) = 4x - 4$ when $x > 2$ (chosen so f is continuous at $x = 2$), determine whether f is differentiable at $x = 2$.

a. the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 4 — they agree, so f is differentiable at $x = 2$. ✓

b. the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 4 — they agree, but f is not differentiable since the pieces use different formulas.

c. the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 4 — they disagree, so f is not differentiable at $x = 2$.

d. f is differentiable at $x = 2$ automatically, since it is continuous there.

Continuity is given. For differentiability, compare one-sided derivatives: the left-hand derivative is $2(2) = 4$ and the right-hand derivative is 4 — they agree, so f is differentiable at $x = 2$.